Mahmoud Alyosify

MLOps & Cloud AI Engineer

New Cairo, Egypt · Open to relocation and remote · +20 114 555 7959 · mahmoud.alyosify@gmail.com
linkedin.com/in/mahmoudalyosify · github.com/MahmoudAlyosify · mahmoudalyosifysite.github.io

SUMMARY

MLOps and cloud AI engineer who takes trained models to **reliable, affordable services**. Packaged models with FastAPI, ONNX and Docker, deployed a fine-tuned LLM on AWS, and ran my thesis as reproducible benchmarks on an NVIDIA A40. MSc in Artificial Intelligence, Queen's University, Canada (GPA 4.30 / 4.30); three AWS Academy credentials.

SKILLS

Cloud & DevOps: AWS, Docker, GitHub Actions, Linux, Bash
Serving: FastAPI, ONNX, FAISS, streaming APIs, caching
Reliability: automated testing (pytest), monitoring (Prometheus), reproducible pipelines
ML & data: PyTorch, scikit-learn, Hugging Face, LangChain, SQL, PySpark, Pandas

EXPERIENCE

AI Engineer Intern — Injaz Tech (Injaz Group), Cairo — Oct 2026 – Present

- Wrote the **Docker** setup and a **GitHub Actions** workflow (linting, 136 tests, evaluation gate); added health checks, Prometheus metrics, retries and a circuit breaker.
- Raised top-1 retrieval accuracy from **75% to 94%** and cut false refusals from 5 in 20 to 1 in 20.

Systems Engineer (Military Service) — National Company for Roads Building & Development, Cairo — Feb 2024 – Mar 2025

- Supported the backend systems of Egypt's national toll network, used by **more than half of the country's vehicles**.
- Debugged live transportation software and supported legacy applications to keep downtime to a minimum.

Part-Time Machine Learning Instructor — National Telecommunication Institute (NTI), Egypt — Jul 2025 – Present

- Teach **90-hour machine learning bootcamps** and AI for Business modules in NTI's Summer Internships Program.

Online Instructor — Udemy & YouTube (Arabic courses), Remote — 2020 – Present

- Created Arabic courses on algorithms, discrete mathematics, probability and statistics: **10,000+ learners, 700K+ views**.

PROJECTS

Horus-OSINT — fine-tuned LLM on AWS (Llama-3-8B, QLoRA, AWS) — 2026

- Fine-tuned **Meta-Llama-3-8B** on **159,826** global event and terrorism records (GTD, GDELT) and deployed it on **AWS** as an assistant that answers questions in plain language.

RL-Cloud-Autoscaler — deep RL for cloud scaling (Python, Gymnasium, PPO, DQN) — 2026

- Built a cloud simulation (Gymnasium) and trained **PPO** and **DQN** agents to balance cost and latency, compared with standard threshold-based autoscaling.

MinimalLM — Master's thesis (Python, PyTorch, Hugging Face Transformers) — 2026

- Built to be reproduced: pinned models and GPU, greedy decoding, 56 automated tests, and a notebook that recomputes every number from the stored answers.

SimCLR-Vision-SSL — self-supervised image model (PyTorch, ONNX, FAISS) — 2026

- Exported the model to **ONNX** and served image search through a **FAISS** index, ready to deploy.

EDUCATION

MSc in Artificial Intelligence — Queen's University, Kingston, Canada — Sep 2025 – Oct 2026

- GPA **4.30 / 4.30** · Completed October 2026 · Digilians Presidential Scholarship

BSc in Computer & Information Science (Bioinformatics) — Assiut University, Egypt — Sep 2019 – Jul 2023

- GPA 3.53 / 4.00 (Very Good with Honours) · Graduation project graded A+

CERTIFICATIONS & AWARDS

AWS Academy: ML Foundations, ML for NLP, Data Engineering · IBM AI Developer Professional Certificate (IBM, 2026) · Deep Learning Specialization (DeepLearning.AI, 2026) · 1st Place, Smart Cities Hackathon 2022 (77 university teams)

LANGUAGES

Arabic (native) · English (professional working proficiency)